

Tile Editor User Manual

Terrain and map editor for Railroader mods

Contents

1. Getting Started
2. Keybind Reference
3. Terrain Editing
4. Track Editing
5. Mod Tools
6. In-Game Geo Workspace

Getting Started

The Tile Editor is a desktop application for building Railroader maps: terrain tiles, track graphs, roads and rivers, scenery, operations, signals, and crossings. It pairs with an in-game F9 workspace that edits the same data live.

Requirements

- 64-bit **Python 3.10 or newer**
- pygame-ce, numpy, Pillow, requests, scipy

```
pip install -r requirements.txt
```

Running

Windows:

```
run_editor.bat
```

The launcher finds Python through the py launcher, PATH, registry entries, and common python.org, Conda, and Scoop locations — **Python does not need to be on PATH**. If you use a portable or unusual install, set `TILE_EDITOR_PYTHON` to the exact `python.exe` path.

Any platform:

```
python -m edit_tiles
```

The Interface

A two-row navigation bar sits at the top: **row 1 is terrain tools, row 2 is mod tools**. The minimap is top-right, the status bar is at the bottom, and selection properties appear top-left when something is selected.

Press `?` or **F1** at any time for nine pages of in-app help. Full bindings are in `KEYBINDS.md`.

First Session: Editing Track

- 1 **Mod** → **Open Mod Folder** and pick a mod folder (it needs a `Definition.json`).
- 2 The map shows every track layer, coloured by source file.
- 3 **Click a node** — its properties appear top-left.
- 4 **Drag a selected node** to move it. Terrain height is sampled automatically on release.
- 5 **Ctrl+click** empty space (or Geo → Add Node) to place a new node.
- 6 Select a node, click **Connect** →, then Ctrl+click a second node to draw a segment.
- 7 Changes auto-save to `game-graph.json` and hot-reload into a running game in about half a second.

First Session: Editing Terrain

- 1 Load tiles — **Load Tiles**, or drag a tile folder onto the window.
- 2 Press F to fit them in view.
- 3 Press E for edit mode. The cursor becomes a brush.

- 4 **LMB paints, RMB erases.** [and] size the brush; - and = change strength.
- 5 Press B to cycle brush type — Raise, Flatten, Paint, Smooth, Noise, Erode.
- 6 **Ctrl+S** saves modified tiles. **Ctrl+Z** undoes.

Press D for diff mode at any point to see which tiles you have changed.

Display Modes

Key	Mode
H	Heightmap — elevation contours, dark green low to yellow/white high
V	Vegetation — dominant biome per pixel (presets 0–7)
W	Water — white is water, black is land
D	Diff — tiles modified since the last save

S toggles hillshade, which adds directional lighting and is the fastest way to read terrain shape.

Live Connection To The Game

The ●LIVE/OFF indicator in the nav bar shows the TrackBridge connection.

With `Hogers.TileEditorBridge` installed in Railroader, edits made in the desktop editor hot-reload into the running game, and the in-game **F9** workspace edits the same data back. Track, scenery, splineys, telegraph poles, and terrain tiles synchronise in both directions.

Dirty-edit ownership locks prevent both sides writing at once. If a reload arrives while you have unsaved work, it is preserved as a timestamped conflict copy rather than being overwritten.

See In-Game Geo Workspace.

Interface Scale

Ctrl+- and Ctrl+= shrink and grow the whole UI; Ctrl+0 resets to 100%. The setting persists between launches. The sidebar A- / 100% / A+ buttons under Workspace do the same.

The Reload Button

Reopens the current track graph and mod JSON from disk. Its colour tells you the state:

Colour	Meaning
Grey	Nothing loaded that can be reloaded
Blue	Reload available
Yellow	Reload available and unsaved changes would be discarded

Undo

Ctrl+Z covers terrain strokes and mod edits, up to 50 mod steps. The status bar shows ■ N with the number available.

Batch tile deletion is also undoable — deleted tiles move to a timestamped `_TileEditor_Deleted_Tiles` recovery folder, and `Ctrl+Z` restores the whole batch.

Next

- Keybind Reference — every binding
- Terrain Editing
- Track Editing
- Mod Tools
- Data Formats

Keybind Reference

Every keyboard and mouse binding in the desktop Tile Editor. Press ? or **F1** in the editor to see the same table in-app (help page 9).

Bindings are **contextual** — several keys do different things depending on whether edit mode is on. Those are marked.

Cheat Sheet

The dozen you will actually use every session:

Key	Action
E	Toggle edit mode (terrain painting)
F	Fit all tiles in view
Ctrl+S	Save
Ctrl+Z	Undo
Ctrl+click	Create node at cursor
RMB drag	Pan the map
Scroll	Zoom
[]	Brush smaller / larger
- =	Brush weaker / stronger
T	Toggle track overlay
Esc	Cancel anything / close panel
? / F1	Help

Display Modes

Key	Action
H	Heightmap mode — elevation contours
V	Vegetation mode — veg/biome layer
W	Water mode — water coverage
S	Toggle hillshade
D	Toggle diff overlay (modified tiles)
O	Toggle OSM overlay
T	Toggle track display
N	Toggle node dots
I	Toggle tile info tooltip

View And Navigation

Key	Action
F	Fit view to tiles
R	Redraw / refresh
Ctrl+R	Reload graph and mod data from disk
Scroll	Zoom in / out
RMB drag	Pan the map
MMB	Pan (middle mouse)
Mousewheel over panel	Scroll lists
← → Tab	Navigate help pages

Note: help page 2 lists LMB drag for panning while the keybind summary lists RMB drag. Right mouse is the reliable pan outside edit mode; in edit mode LMB paints, so use RMB or MMB there.

Interface Scale

Key	Action
Ctrl+-	Shrink UI scale
Ctrl+=	Grow UI scale
Ctrl+0	Reset UI scale to 100%

Scale persists between launches. The sidebar A- / 100% / A+ controls under Workspace do the same thing.

Files

Key	Action
Ctrl+S	Save modified tiles / mod project
Ctrl+Z	Undo — terrain strokes or mod edits (up to 50 mod steps)
L	Load track graph JSON
B	Pick game / bridge folder (<i>outside edit mode</i>)
Drag-drop folder	Drop a tile folder on the window to load it
Esc / Q	Quit, cancel, or close

Terrain Editing

Edit mode only — press E first. LMB paints, RMB erases.

Key	Action
E	Toggle edit mode
B	Cycle brush type (<i>edit mode</i>)

Key	Action
[/]	Brush size down / up
Ctrl+Scroll	Brush size
- / =	Brush strength down / up
LMB	Paint / raise
RMB	Erase / lower
MMB	Eyedropper — sample height or veg at cursor
0–7	Vegetation preset (<i>veg mode</i>)
,	Set height floor clamp to cursor height
.	Set height ceiling clamp to cursor height
Ctrl+,	Clear floor clamp
Ctrl+.	Clear ceiling clamp
Ctrl+X	Export heightmap as PNG

Brush Types

Cycle with B in edit mode.

Brush	Effect
Raise	Raise (LMB) or lower (RMB)
Flatten	Level to the height you first clicked
Paint	Stamp an exact target height
Smooth	Average pixels with neighbours
Noise	Add fBm procedural noise
Erode	Simulate hydraulic erosion

Terrain Selection

Press M in edit mode.

Key	Action
M	Toggle selection mode
Click+drag	Rect / lasso select
Click (wand)	Flood-fill select similar values
Ctrl+C	Copy selection
Ctrl+V	Paste selection — click to place
Ctrl+X	Cut selection / export PNG
Delete	Clear selected region

Track Editing

Input	Action
LMB	Select node or segment
Click segment	Selects anywhere along the line, not just the midpoint
Click empty space	Deselect
Click selected node	Second click starts drag
Drag release	Commit move — samples terrain height
Ctrl+click map	Create node at cursor
Ctrl+click node	Finish a connection to that node
Delete	Delete selected node or segment
Esc	Cancel drag / connect / place
Ctrl+drag	Rubber-band select nodes
Shift+drag	Add to group selection

Connection Feedback Colours

Drag a node toward a target and watch the highlight — it tells you what will happen on release:

Highlight	Result
Snap ring on a node	Connect the two with a segment
Cyan on a segment	Insert the node into that segment (splits it in two)
Yellow on a segment (Shift+drag)	Insert the node and add a turnout diverge leg

Inserting into a segment deletes the original and creates two; the new node inherits a bezier heading. The yellow turnout variant uses the current Geo → Turnout settings for angle and length.

Tile Generation Panel

Input	Action
Click tile cell	Queue a tile for generation
Drag cells	Box-select a region to queue
RMB tile cell	Remove from queue
Scroll	Zoom the generate grid
MMB drag	Pan the generate grid
Run	Generate all queued tiles

Contextual Keys

Three keys change meaning with edit mode, which is the most common source of confusion:

Key	Outside edit mode	In edit mode
B	Pick game / bridge folder	Cycle brush type
LMB	Select / pan	Paint
RMB	Pan	Erase

M, [,], -, =, ,, ., and 0–7 only do anything in edit mode.

Status Bar

The bottom bar reports the current action and errors. ■ N shows how many mod undo steps are available.

Related

- [Terrain Editing](#)
- [Track Editing](#)
- [Getting Started](#)

Terrain Editing

Sculpting heightmaps, painting vegetation and water, and generating tiles.

Press E to toggle edit mode. The cursor becomes a brush and a toolbar appears below the nav bar. **LMB paints**, **RMB erases**.

Brushes

Cycle with B in edit mode.

Brush	Effect
Raise	Raise (LMB) or lower (RMB)
Flatten	Level the area to the height you first clicked
Paint	Stamp an exact target height
Smooth	Average pixels with their neighbours
Noise	Add fBm procedural noise
Erode	Simulate hydraulic erosion

Flatten samples on first click, so click on the elevation you want *before* dragging across the area you want levelled.

Brush Controls

Key	Action
[/]	Size down / up
Ctrl+Scroll	Size
- / =	Strength down / up
MMB	Eyedropper — sample height or vegetation at the cursor

Height Clamps

Clamps bound what a brush can do, which is how you grade against a fixed elevation without overshooting.

Key	Action
,	Set floor clamp to cursor height
.	Set ceiling clamp to cursor height
Ctrl+,	Clear floor clamp
Ctrl+.	Clear ceiling clamp

Set a floor at rail height and Smooth aggressively around a right-of-way — the clamp stops the brush pulling terrain below the grade.

Vegetation And Water

Press **V** for vegetation mode, **W** for water.

In vegetation mode, keys 0–7 select the preset to paint. Water mode paints the water mask — white is water, black is land.

Selection

Press **M** in edit mode.

Tool	Use
Rect / Lasso	Click and drag to select a region
Wand	Click to flood-fill select similar values

Key	Action
Ctrl+C	Copy
Ctrl+V	Paste — then click to place
Ctrl+X	Cut / export PNG
Delete	Clear the selected region

Copy and paste move terrain between locations, which is the quickest way to repeat a landform.

Tile Generation

Open the **Generate** panel to build terrain from real-world data.

Action	Result
Click a tile cell	Queue it
Drag across cells	Box-select a region to queue
RMB a cell	Remove from the queue
Scroll	Zoom the grid
MMB drag	Pan the grid
Run	Generate everything queued

Queue the whole area first, then Run once — generation is the slow part.

Tile Cleanup

A dedicated workspace for trimming large generated blocks down to what a railroad actually needs.

Action	Result
Drag	Mark tiles
Shift+drag	Add to the marked set
Ctrl/right-drag	Keep tiles

Action	Result
Invert / Outside ROW	Mark everything outside the right-of-way

The usual workflow is to mark the ROW, then **Invert / Outside ROW**, then delete.

Batch deletion asks for confirmation and moves files into a timestamped `_TileEditor_Deleted_Tiles` recovery folder. `Ctrl+Z` **restores the whole batch** — nothing is destroyed immediately.

OSM Guide Overlay

Press `O` for the OpenStreetMap overlay, which is the practical way to trace real alignments.

The in-game `F9` equivalent streams an aligned guide over just the 5×5 or 8×8 game-tile neighbourhood around the camera, following terrain and unloading behind you. It shares the desktop editor's `Map.json` georeference and caches images under the installed mod rather than a machine-specific path.

Setting	Options
Mode	Clear Lines (default — drops the pale map sheet, keeps features) or Full Map
Resolution	Overview → Ultra, matching <code>z15–z18</code>

Adaptive detail keeps the camera area sharp without rendering the whole coverage window at maximum resolution. Both the desktop OSM toolbar and the `F9` Terrain controls show cache usage and offer a confirmation-protected **Clear Cache**.

OSM and Mapbox data come from third-party services under their own terms — check those before redistributing anything derived from them.

In-Game Terrain Workspace

The `F9` Terrain workspace separates practical sculpting from vegetation and water painting. Its building-pad, track/road, walkway, grade-plane, ditch, and embankment tools use bounded cut/fill with target convergence, specifically to avoid the spikes a naive flatten produces.

Saving

`Ctrl+S` saves all modified tiles. `Ctrl+X` exports the heightmap as a PNG. Press `D` for diff mode to see exactly which tiles are unsaved — they carry a yellow border.

Related

- [Keybinds](#)
- [Getting Started](#)
- [Track Editing](#)

Track Editing

Building and editing track graphs on real terrain.

Every change auto-saves to `game-graph.json` and hot-reloads into a running game in roughly 500 ms.

Selecting

Action	Result
Click a node or segment	Select — properties appear top-left
Click a segment	Works anywhere along the line , not just the midpoint
Click empty space	Deselect
Click an already-selected node	Second click starts drag mode
Release a drag	Commits the move and samples terrain height
Esc	Cancel drag, connect, or place
Delete	Delete the selected node, or all its segments

The two-click drag is deliberate: one click selects, a second starts the move, so you cannot nudge a node by accident while inspecting it.

Node Properties

Control	What it does
X / Y / Z	Direct position editing — click, type, Enter
RotY nudge	Fine grid: $\pm 90 / 45 / 30 / 15 / 10 / 5 / 1 / 0.1 / 0.05 / 0.001^\circ$
Flatten	Zeros rotX and rotZ, levelling the node on terrain
Reverse	Flips heading 180°
Merge	Removes a middle node between exactly two segments
Split	Duplicates the node and rewires the selected segment to the copy
Copy XYZ	Copies position to the clipboard
Paste Y	Pastes only the height onto the node

Paste Y is the tool for levelling a run: copy from a node at the right elevation, then paste height onto each node along the tangent without disturbing plan position.

The connected-segments list at the bottom of the node panel is clickable.

Segment Properties

Control	What it does
Track Class	Mainline / Branch / Industrial — coloured buttons
Style	Standard / Yard / Bridge / Tunnel
Speed	Nudge $\pm 25 / 10 / 5 / 1$ mph, or type a value

Control	What it does
Priority	Editable box
GroupID	Track group membership
Reverse	Swaps startId and endId
Trestle	Wraps the segment in an AutoTrestleBuilder spliney

A class button brightens when that class is active.

Creating Track

Action	Result
Ctrl+click map	Create a node at the cursor, sampling terrain Y
Geo → Add Node	Placement mode with a live-coordinate crosshair
Connect → button	Enter connect mode from the selected node
Ctrl+click a node	Finish the connection
Drag node → node	Snap ring; release to connect
Drag node → segment	Cyan ; release to insert into that segment
Shift+drag → segment	Yellow ; insert and add a turnout diverge leg

New nodes default to Mainline, Standard, 45 mph, and their rotY is set automatically to face the connection direction.

Inserting Into A Segment

Dropping a node onto a segment deletes the original and creates two, giving the new node a bezier heading so the alignment stays smooth. The Shift variant does the same and adds a turnout diverge leg using the current **Geo** → **Turnout** settings for angle and length.

For a full switch with proper geometry, use **Geo** → **Turnout**: select the node, Preview, then Commit.

Group Operations

Action	Result
Ctrl+drag	Rubber-band select nodes
Shift+rubber-band	Add to the existing selection
Apply dX / dY / dZ / Rot	Bulk translate and rotate the selection

Spliney Control Points

Rivers, roads, and trestles are splines with control points — the yellow dots. Zoom in to click one.

Control	What it does
Click, click again	Select, then start dragging

Control	What it does
Drag release	Moves the point; terrain can resample Y
Prev / Next	Step along the spline
Ins Before / Ins After	Insert a control point
Sample Y	Re-read terrain height
Auto Rot	Rebuild rotY from neighbouring points
Delete Pt	Remove a point (the spline keeps at least two)
Fit Terrain	Terrain-fit the entire road or river
Reverse Flow	Rivers only — reverses points and adds 180° rotY

River preview arrows show flow from the first point to the last, so check them after a Reverse Flow.

Gauge

Segments carry a gauge field consumed by FUSE Narrow Gauge:

Standard (default) · Narrow · DualGauge · DualGauge_L · DualGauge_R · DualGauge_T

New arcs, pieces, parallel tracks, turnouts, wyes, and direct connections inherit the active gauge. Existing gauge and companion fields survive splits, merges, and renames.

RailLoader ignores the gauge field, so one graph works for both runtimes. Visible narrow and dual rail geometry is rendered by FUSE Narrow Gauge, not by the Tile Editor — the field is metadata.

DualGauge_T must be **one short segment** between opposite explicit L and R runs. See Data Formats.

Geometry Tools (Geo)

The Geo panel carries seven tabs — Spliney, Arc, Parallel, Fit Arc, Turnout, Wye, Span, and Turntable, alongside grade work and piece-based assembly.

Guide geometry stays draft-only until you fit an arc or build a spliney, so you can trace freely without committing anything.

Calculators (Calc)

Calculator	Input → output
Crossover	Separation + angle → leg lengths
Curved Turnout	Radius + gauge + angle → diverge geometry
Grade / Slope	Run + rise → percent, ratio, angle
Measure	Run, rise, grade, and heading between two picks

Related

- Keybinds
- Mod Tools
- Data Formats

Mod Tools

The second nav row. These panels edit everything in a map mod that isn't terrain or raw track geometry.

Every panel writes to the owning mod JSON layer and triggers a bridge reload. `Ctrl+Z` undoes panel edits too, up to 50 steps.

Mod Panel

Control	What it does
Open Mod Folder	Load a mod folder (needs <code>Definition.json</code>)
Open Base Game	Load a standalone <code>graph-data.json</code>
New Mod	Create a new mod folder structure
Save All	Save every dirty layer file
Layer dot	Toggle layer visibility
Layer row	Set as the active layer
● indicator	Unsaved changes in that layer

Columns show per-layer counts: Nodes (plus deleted), Segs, Spl (splineys), Areas.

Closing the panel with `X` or `Esc` leaves the mod loaded.

Map Legend

What you see with a mod loaded:

Appearance	Meaning
Yellow lines	<code>game-graph.json</code> track segments
Grey lines	Base game track
Blue polylines	Rivers (FlowyThingBuilder)
Tan polylines	Roads (FlowyThingBuilder)
Grey/cream	AutoTrestle bridge spans
Orange circles	Turntables
Green diamonds	Stations
Orange diamonds	Loaders / industry
Coloured squares	Scenery buildings — zoom in for the model name
Large circles	Area/town centres with name labels
Orange circle	Live locomotive (from the bridge)
Teal square	Live railcar (from the bridge)

The last two only appear with a live game connected.

Progression Editor (Prog)

Left column lists sections in topological order, prerequisites first. Right column lists map features and what each unlocks.

Control	What it does
+ Section	Add a purchasable section
+ Feature	Add a map feature
Del Section / Feature	Remove the selected item
Save	Write back to <code>progressions.json</code>

Section fields: id, display name, prerequisites, cost, feature to enable. Feature fields: id, display name, areas unlocked, track groups.

Topological ordering means a section always appears after everything it depends on — if one is in an unexpected place, its prerequisites are the reason.

Area / Town Editor (Areas)

Three columns: areas (sorted by order, with a source-layer colour dot), industries in the selected town, and components in the selected industry.

Control	What it does
Click a row	Select and load the next column
Go to Area	Pan the map there and close the panel
+ Area	Create at the current map centre
+ Industry	Create in the selected town
+ Component	Create in the selected industry
Edit Area / Industry / Comp	Edit that JSON object directly
Del Area	Mark for deletion (not saved yet)
Del Industry / Comp	Remove immediately from its parent
Save	Write all dirty town JSON files

Note the asymmetry: deleting an area only marks it, while deleting an industry or component removes it from the parent right away.

Industry creation exposes the full RailLoader/FUSE component set — storage capacity and daily change, car transfer rates and ordering, formula input/output maps, team-track import/export profiles, passenger population and neighbours, repair/overhaul, interchange variants, progression, multiple TrackSpans, and custom typed fields.

Scenery Placement (Scenery)

Control	What it does
Model ID field	Type an identifier or click a quick-pick
RotY nudge	± 90 / 45 before placing

Control	What it does
Place on Map	Placement mode — click to drop
Esc	Exit placement mode
Go To	Pan to the selected object
Del Object	Remove from the layer

Y position is auto-sampled from terrain at the click point.

The in-game Scenery workspace adds a searchable loaded-asset palette that includes runtime-resolvable RailLoader SCAssetPacks registered after the initial catalog, plus in-world picking, transform controls, terrain snapping, and duplication.

Mandela / Prefab Instances (Mandela)

Places instances of base-game prefabs.

Control	What it does
Target	Destination scene path key
Prefab	Base-game prefab path to instantiate
Base Pick	Search dumped-mandelas.txt and fill Prefab
Reload Base	Reload the dumped prefab catalog
Load Sel / Save Sel	Pull draft values from an entry, or write them back
Duplicate	Copy a placement to a new target path
Place on Map	Drop the draft at terrain height
Enabled	Cycle default / enabled / disabled

FUSE imports the same entries as world.sceneClones, so one graph stays compatible with both runtimes.

Objects Workspace (in-game)

Selects base-game buildings and props under the mouse and saves move, rotate, scale, enable/disable, and safe-clone operations as portable RailLoader mandelas.

Shared town, map, and world roots are blocked, so clicking one station cannot move the entire scene. Thin signs and small props get a screen-space picking halo when the ray misses.

The F9 OBJECTS page also identifies Railroader's original scene signs and gives them a dedicated visible/hidden toggle, without treating loader-added scenery signs as base-game objects.

Track Spans (Spans)

Lists all spans across layers with their layer colour. Click a span to edit its upper and lower segment, distance, and end. + **New Span** and **Del** manage entries.

Spans may cover a whole segment, a measured partial range, or connected start/end segments.

Group Move (Group)

Ctrl+drag rubber-bands a node selection; Shift+rubber-band adds to it. Apply dX / dY / dZ / Rot to translate and rotate everything selected at once.

Calculators (Calc)

Calculator	Input → output
Crossover	Separation + angle → leg lengths
Curved Turnout	Radius + gauge + angle → diverge geometry
Grade / Slope	Run + rise → percent, ratio, angle
Measure	Run, rise, grade, heading between picks

Spliney Panel

Control	What it does
Type	Road or River
Target	An existing writable spliney layer
New JSON...	Create and register a new road or river file
Width / Seed	Defaults for new placements
Use Sel Rot	Use the selected track node's heading
Place on Map	First click creates a spliney with two starter points
Go To / Delete	Navigate to or remove the selected spliney

The details card shows style, width, point count, and length.

Loaded base-game roads and rivers also expose their real control nodes. The first change creates a same-name override in the selected graph, and every adjustment refreshes the associated height, roadbed, dirt, object, water, and vegetation-mask terrain modifiers.

Telegraph Poles (in-game)

Amber world markers, pole creation at the pointer, continuous-line placement, manual wire connections, WORLD/LOCAL movement, full pitch/heading/roll rotation, and deletion.

New poles and edges persist in the map mod's `tile-editor-telegraph-poles.json`. Original poles still save compatible cumulative Alina `TelegraphPoleMover` offsets plus portable rotation overrides.

Related

- Data Formats — what these panels write
- In-Game Geo Workspace
- Track Editing

In-Game Geo Workspace

Hrogers.TileEditorBridge puts a live editing workspace inside Railroader, on top of the game's own Track.Graph. Press **F9**.

It does not use Alina's Map Editor, and it can select and remember an installed RailLoader mod or game-graph by itself — **the desktop Tile Editor does not need to be running**.

Mouse Handling

While F9 is open the mouse is reserved for editing. Railroader's pan, orbit, look, and wheel zoom are locked so the camera stays put during placement.

- **W A S D** and the normal fast-movement modifiers still move the camera.
- **Hold middle mouse** to temporarily restore all normal camera controls. Editing clicks are ignored until you release it.
- **Right-click the world** to clear every selection and cancel any active mode — placement, connection, bridge-picking, Fit Arc, TrackSpan, pole-wiring, terrain, and delete.

That right-click is the universal escape hatch; reach for it when a mode seems stuck.

Workspaces

Workspace	Covers
Geo	Nodes, segments, grades, pieces, arcs, parallel track, fitted arcs, turnouts, wyes, splineys
Terrain	Sculpting, plus vegetation and water painting, and the OSM guide
Scenery	Searchable asset palette, placement, transforms, terrain snapping
Objects	Base-game buildings and props → portable mandelas
Poles	Telegraph pole nodes and wire edges
Signals	Semaphore placement and interlockings
Operations	Towns, spans, industries, stations, loaders, turntables
Desktop	Editor status; remotely focus, undo, save, and reload

The panel is resizable.

Track Editing

A fast shortcut for connected track: **click the first node normally, then Shift-click the second** to connect them. The destination stays selected, so you can keep Shift-clicking to build a chain.

Action	Result
Place Free Node	Create and select an independent starting node anywhere on terrain
Add +10 m	Create a connected node ahead using current heading, grade, and bank
Ctrl+drag a node	Move it over terrain as one undoable edit
Drop over a cyan node	Connect the two, keeping the dragged node at its last valid position

Repeated **Add +10 m** clicks extend track continuously, selecting each new endpoint — the quickest way to run a tangent.

Node transforms live in their own movable, resizable child window. The main Geo surface keeps a compact summary and continue-track controls, while naming, move/rotate, exact coordinates, actions, and selective copy/paste stay together in the focused Node Editor.

Its direction pad separates elevation from WORLD/LOCAL plan movement and groups pitch, heading, and roll into paired curved-arrow controls, with hover text on every axis.

Property Clipboard

Copy a single field rather than the whole node: elevation, grade, heading, bank, full rotation, an elevation+rotation combination, the turnout switch flag, or everything. Only compatible paste actions light up on the target. X/Z position and connections are never changed.

Naming

New nodes can use a remembered prefix and readable name pattern. Every builder shares the pattern and adds a collision-safe number.

Gauge

Track creation and segment properties support Standard, three-foot Narrow, automatic DualGauge, explicit DualGauge_L / DualGauge_R, and DualGauge_T transitions. Track class can be changed live between Mainline, Branch, and Industrial with undo/redo and schema-safe output.

F9 distinguishes saved gauge metadata from a loaded live runtime, tells you when FUSE and FUSE Narrow Gauge must be enabled before restarting, and offers a live gauge synchronisation action when both are active.

Performance detail worth knowing: ordinary 3-foot node and segment edits batch their FUSE metadata and rebuild only affected endpoints. **Dual-gauge topology edits are deferred and coalesced**, because their generated ghost and shared rails need the complete synchroniser — so a dual-gauge change costs more than a narrow one.

DualGauge_T is authored as one short segment between opposite explicit L/R runs; F9 checks its two endpoints and prevents applying it through a whole chain.

Signals

Places Railroader's base-game animated semaphore assemblies with one, two, or three heads: pointer placement, amber world selection, WORLD/LOCAL movement, full rotation and flip controls, exact transforms, and test aspects.

Masts support a one-time rail snap or a **persistent track lock**. A locked mast keeps its side, height, and facing offsets and follows later Bezier curve, grade, and elevation edits.

The **Diamond Interlocking** builder detects the real intersection of two non-connecting Bezier segments, rejects a grade-separated overpass, and places four independently adjustable A1/A2/B1/B2 semaphores. Setback, lateral and vertical offset, head count, approach locking, and release length are all configurable. Long 500–800 m setbacks follow connected track across segment boundaries automatically.

Output is portable `train-signals.json`, loaded during ordinary gameplay by Railroad Operations — players never install the Tile Editor. See Data Formats.

Live **Signals & CTC**, **Train Orders**, and **My Orders** desks join the normal Company → Operations window. F9 remains the map-authoring editor.

Splineys And Bridges

The in-game Spliney workspace can build a bridge directly from a clicked track segment. Its two endpoint controls inherit the track node positions and rotations, reproducing Railroader's exact 3D Bezier span with an adjustable below-rail deck offset and no redundant intermediate nodes. Yellow track picking stays armed for the next bridge afterwards.

Road, river, bridge, and trestle control points show movement plus full pitch/heading/roll together, using the same arrow controls and precision steps as track nodes and scenery.

Operations

Discovers, searches, selects, and edits towns, TrackSpans, industries, passenger stops, freight components, engine facilities, physical coal/water/fuel loaders, custom loader prefabs, station agents, commodities, and turntables. Pointer placement and coloured world overlays use the same undoable document workflow as track and scenery.

Dedicated Geo **Span** and **Turntable** tools sit beside Arc, Turnout, and Wye. Turntables support native FUSE pit and bridge geometry, bridge-track gauge, subdivisions, and optional roundhouse stalls, plus preserved legacy RailLoader/Alina TurntableBuilder output. Standard 30 m and three-foot narrow-gauge presets are included.

Sync With The Desktop Editor

Track, scenery, road/river/bridge splineys, telegraph poles, and terrain tiles synchronise in **both directions**.

Dirty-edit ownership locks prevent simultaneous writes. An incoming reload preserves unsaved work as a timestamped conflict copy rather than discarding it.

Mods/TrackBridge files carry the live graph and reload traffic; the compact UMM panel adds a small status and request channel beside them without changing that format.

Rendering Notes

Yellow segment overlays live in a protected graph-level editor layer keyed by segment id, so Railroader track rebuilds and undo/redo no longer permanently remove editable track lines.

Overlays share lightweight materials, cache repeated asset searches, reduce long-segment pick markers, and sleep distant track visuals and colliders until the camera approaches.

Related

- Data Formats
- Mod Tools
- Getting Started